Which Capacity Matters Most for Earned Revenue? Evidence from U.S. Nonprofit Performing Arts Organizations

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Nonprofit performing arts organizations have relied on government support and private donations, both inherently unstable due to political priorities and macroeconomic fluctuations. Earned revenue has been proposed as a strategy for financial sustainability, and prior research emphasizes organizational capacity as the key factor enabling it. Yet capacity is multidimensional, and because policy resources are limited, it is necessary to know which dimension to prioritize. Using SMU DataArts Cultural Data Profile panel data (2019–2024) and a two-way fixed effects model, this study examines how human, programmatic, and financial capacity relate to the earned revenue ratio among U.S. nonprofit performing arts organizations (1,864 organizations; 7,489 organization-year observations). Human and programmatic capacity show significant positive associations, with human capacity emerging as the strongest correlate; financial capacity, contrary to expectations, shows a negative association. These findings offer empirical grounds for prioritizing among capacity dimensions in cultural policy design.

Keywords: cultural policy; nonprofit performing arts organizations; earned revenue; organizational capacity; financial sustainability; nonprofit finance

Introduction

Nonprofit performing arts organizations are inherently unable to sustain operations without external funding. According to Baumol and Bowen's (1966) cost disease thesis, the performing arts are a labor-intensive industry with limited scope for productivity growth, making it difficult for ticket revenue alone to keep pace with rising operating costs. Government subsidies and private donations have historically compensated for this structural limitation.

However, the external funding on which performing arts organizations depend is itself unstable, shaped less by organizational performance than by political conditions and macroeconomic cycles beyond organizations' control—from recurring threats to eliminate the National Endowment for the Arts (NEA) budget to the long-term decline in the real share of government support (Felton, 1994). Against this backdrop, expanding earned revenue has been proposed as a strategy for strengthening financial sustainability (Dees, 1998). Yet as McCarthy et al. (2001) point out, the share of earned revenue has remained largely unchanged over recent decades even as total revenue has steadily grown—meaning that while the importance of earned revenue is widely recognized, empirical evidence on the organizational conditions needed to actually expand it remains limited.

Prior research attributes the expansion of earned revenue to organizational capacity (Hall et al., 2003), since generating revenue directly—unlike receiving external funding—presupposes the capacity to design and operate marketable programs and to build relationships with audiences. Yet current funding structures are often project-based, and support for capacity-related costs such as personnel has remained persistently constrained (Gregory & Howard, 2009). This raises an important policy question: if supporting organizational capacity is necessary to strengthen financial sustainability, which capacity should policy prioritize? Policy resources are limited, and supporting all forms of organizational capacity is not realistic. Yet few studies have jointly examined these dimensions to assess whether they matter equally for earned revenue generation: prior research has typically examined human, programmatic, and financial capacity separately, rather than comparing them within the same analytical framework. This study addresses this gap by examining which capacity dimension is most closely associated with earned revenue. Specifically, it is hypothesized that human capacity (H1), programmatic capacity (H2), and financial capacity (H3) are each positively associated with the earned revenue ratio. Using SMU DataArts Cultural Data Profile (CDP) national panel data from 2019–2024, a two-way fixed effects model is applied to estimate these relationships.

This study makes two contributions. First, it addresses a methodological gap by comparing human, programmatic, and financial capacity simultaneously within a single fixed-effects framework. Second, it translates this comparison into an empirical basis for identifying which dimensions warrant priority under resource constraints—a question that, as the findings below show, does not have a straightforward answer.

Literature Review

2-1. The Structural Inevitability of Dependence on External Resources

The starting point for understanding the financial structure of nonprofit performing arts organizations is Baumol and Bowen's (1966) "cost disease" thesis. In labor-intensive industries such as the performing arts, productivity gains are inherently limited, while wages continue to rise in line with the broader economy. As a result, the cost of producing performances increases over time without corresponding gains in productivity. Because ticket prices and other earned revenue sources cannot expand indefinitely to offset these rising costs, nonprofit arts organizations face a persistent structural gap between revenues and expenditures—often referred to as the "earnings gap."

McCarthy et al. (2001) provide empirical evidence of how this structural constraint plays out in practice. Between 1977 and 1997, the average total revenue of nonprofit performing arts organizations increased steadily, yet the share of earned income remained stable despite active marketing efforts and substantial increases in ticket prices. These findings suggest that organizations were unable to offset rising costs through earned income alone and therefore continued to rely heavily on external resources, including donations and government support.

This pattern has persisted well into the twenty-first century. Using CDP data from 2008 to 2016, Liu and Kim (2022) similarly found that private, fee-based revenue accounted for only about 24% of the average arts and cultural nonprofit's total income, with donations (about 44%), government support (about 11%), and other revenue sources accounting for the rest. Together, these findings indicate that the structural earnings gap identified by Baumol and Bowen (1966) and documented by McCarthy et al. (2001) has remained largely unchanged.

2-2. The Structural Instability of External Resources and the Importance of Earned Revenue

The problem is that the two external resources on which nonprofit performing arts organizations depend are themselves exposed to structural instability. Tuckman and Chang (1991), in their analysis of nonprofit financial vulnerability, argue that because nonprofits depend on the munificence of funders rather than on customer satisfaction, they are more vulnerable to revenue instability than organizations based on quid pro quo exchange relationships. Donations and government support are therefore determined less by the quality or performance of an organization's services than by funders' goodwill and political judgment.

Turning first to government support, the NEA has served as the principal federal supporter of the arts since its establishment in 1965 (Hodsoll, 1984), yet its funding has repeatedly been challenged through proposals for substantial budget cuts or even elimination, including during the Reagan and Trump administrations (Lewis & Rushton, 2007; Knight, 2017; Dorgelo, 2025). These recurring political challenges illustrate that public arts funding is contingent rather than guaranteed. Long-term financial evidence reinforces this point. Tracking 25 major U.S. orchestras over 21 years, Felton (1994) found that government support grew by only 7.8% in real terms, compared with 126.6% growth in earned revenue and 158.9% growth in private contributions, while its share of total revenue declined from 11.2% to 5.4%. Likewise, McCarthy et al. (2001) documented declining federal support beginning in the early 1990s, and Liu and Kim (2022) found that government funding still accounted for only about 11% of total revenue among U.S. arts and cultural nonprofits between 2008 and 2016.

Private donations are similarly vulnerable to forces beyond organizations' control. Individual giving declined substantially during the Great Recession and remained depressed for years afterward, failing to recover even as the economy improved (Meer et al., 2017). Corporate philanthropy likewise contracts during periods of economic uncertainty as discretionary expenditures are reduced (Bansal et al., 2015). Together, these studies suggest that donation-based revenue remains unstable because it depends on donor preferences and broader economic conditions rather than organizational performance.

Against this backdrop, earned revenue has attracted growing attention because it is generated through market exchange rather than funders' goodwill or political decisions—the quid pro quo relationship described by Tuckman and Chang (1991). Accordingly, nonprofit organizations have increasingly pursued earned-income strategies—referred to in this study as earned revenue—as traditional funding sources have become more constrained (Dees, 1998).

The importance of earned revenue lies in organizational sustainability itself. Using longitudinal data covering the entire population of registered charities in Canada, Gras and Mendoza-Abarca (2014) show that increasing the share of market-based revenue significantly reduces the risk of organizational exit, although this relationship becomes nonlinear at very high levels of commercial activity. Similar evidence has been found in the U.S. arts and cultural sector, where Liu and Kim (2022) report that earned, fee-based revenue is associated with improved financial health once it exceeds a critical threshold. Together, these findings suggest that earned revenue is not simply an additional source of income but an important organizational resource for long-term sustainability.

2-3. Why Expanding Earned Revenue Requires Internal Capacity

Because earned revenue must be generated by the organization itself, expanding it requires internal organizational capacity. Dees (1998) points out that for nonprofits to generate revenue through commercial activity, they need managerial expertise distinct from general nonprofit operating capacity—such as designing marketable products, marketing, and financial management—and that many nonprofits struggle with commercial activity precisely because they lack this expertise. Honadle (1981) similarly distinguishes between an organization's capacity to attract resources and its capacity to absorb them, noting that being able to secure resources does not mean an organization can use them effectively: even when a grant is obtained, the resource goes to waste without the personnel, skills, and time needed to utilize it. From this perspective, expanding earned revenue likewise presupposes the internal capacity that makes it possible.

However, discussions of what constitutes organizational capacity are diverse, and its definition remains unclear. In a widely cited framework for nonprofit organizations, Hall et al. (2003) conceptualize organizational capacity as comprising three interrelated dimensions: financial capacity, human resources capacity, and structural capacity. Financial capacity refers to an organization's ability to acquire and manage financial resources. Human resources capacity concerns the ability to recruit, develop, and effectively utilize paid staff and volunteers, while structural capacity encompasses the organizational systems, infrastructure, networks, and program-development capabilities that remain beyond individual employees.

The problem is that investment in these capacities is insufficient in practice. According to the "nonprofit starvation cycle" identified by Gregory and Howard (2009), funders underestimate the true costs of running a nonprofit, and organizations, in order to meet these unrealistic expectations, spend less than they actually need—or underreport what they spend—on capacity-related costs such as infrastructure, personnel, and systems, which in turn reinforces funders' distorted expectations in a self-perpetuating cycle. Indeed, while the average overhead ratio across for-profit industries is around 25% of total expenses, a norm of 20% or less has long prevailed in the nonprofit sector, with foundations' indirect-cost allowances averaging only 10–15% of each grant. Lecy and Searing (2015) provide empirical evidence that this pattern is not a temporary phenomenon but a trend that has persisted for decades.

This situation means that the resources available for capacity-building are themselves limited. If resources were unlimited, investing in every dimension of capacity would be the optimal strategy; but because of funders' structural parsimony, the resources organizations can actually devote to capacity-building remain perpetually scarce. Under these conditions, knowing which dimension—financial, human, or structural capacity—should be prioritized for investment to most effectively expand earned revenue carries significant importance.

2-4. The Research Gap: Which Capacity Dimension Is Most Strongly Associated with Earned Revenue?

The question raised at the end of the previous section can thus be restated in operational terms: among human, structural, and financial capacity, which should policy prioritize in order to expand earned revenue?

Yet direct empirical evidence on this question remains scarce. This absence stems in part from a methodological limitation: because the concept of capacity has developed independently across academic disciplines (Christensen & Gazley, 2008), human, structural, and financial capacity have typically been examined separately, using different dependent variables and samples. As a result, prior work has not entered all three into a single model to compare their relative associations with earned revenue—a comparison only possible when each dimension is placed within the same model, sample, and dependent variable.

This gap is compounded within the performing arts specifically. Kerlin and Pollak (2011) show that between 1982 and 2002, commercial revenue rose substantially as a share of total revenue across the nonprofit sector overall, but its share remained essentially flat in arts and culture. This divergence suggests that capacity–revenue relationships established in other subsectors may not extend to the performing arts.

This study takes up this task directly: human, programmatic, and financial capacity are entered into a single two-way fixed effects model, and their relative associations with the earned revenue ratio are compared using national panel data on U.S. nonprofit performing arts organizations from 2019 to 2024.

Data and Methods

3-1. Data

This study uses the national dataset from the Cultural Data Profile (CDP), operated by

SMU DataArts, which aggregates financial and operational information—including revenue sources, expenditures, staffing, and program activity—voluntarily submitted by nonprofit arts and culture organizations. Its administrative panel structure suits this study's focus on within-organization change over time, and it has been used in prior research on nonprofit workforce strategies (Woronkowicz et al., 2020) and revenue composition and financial health (Liu & Kim, 2022).

The analytic sample is restricted to performing arts organizations under NTEE codes A60, A62, A63, A65, A68, A69, A6A, A6B, and A6C. A61 (performing arts centers) was excluded because these organizations primarily function as venue rental and presenter operations rather than producers of performances. The study period covers fiscal years 2019–2024 (six years).

Outlier handling proceeded in three steps. First, observations exceeding the 99th percentile of total_program_occurrences were removed, to prevent a small number of extreme outliers from distorting the distribution even after log transformation. Second, observations with staffing counts exceeding 100,000 were removed; review of the data confirmed these to be clear data-entry errors. Third, extreme values of months_cash falling outside the 1st–99th percentile range were removed.

The full sample consists of 8,113 organization-year observations from 2,485 organizations. Of these, 4 observations lacked data on months of operating cash, and 620 organizations appeared in only a single fiscal year; because organization fixed effects are identified from within-organization variation, which requires at least two observations per organization, these singletons were excluded from estimation. The final estimation sample thus consists of 7,489 observations from 1,864 organizations.

3-2. Variables

The independent variables in this study are constructed to reflect three dimensions of organizational capacity. As discussed above, nonprofit capacity comprises multiple dimensions—financial, human, and structural (Hall et al., 2003)—and this study selects three of these dimensions that are measurable using data and identifiable within a fixed-effects framework: human capacity, programmatic capacity, and financial capacity, corresponding to H1, H2, and H3, respectively.

The dependent variable, earned revenue ratio, is measured as the share of earned revenue in total operating revenue. Measuring this as a ratio rather than an absolute amount follows established practice in the literature, where the share of program-service or earned revenue relative to total revenue—rather than its dollar value—captures the degree to which an organization relies on market-based strategies as opposed to external, contributed resources (Ecer et al., 2017).

Human capacity (H1) is measured as the natural log of total positions (ln_positions). The CDP's total_positions_count_calculation variable sums full-time permanent, full-time seasonal, part-time permanent, and part-time seasonal staff, along with volunteers, independent contractors, interns and apprentices, and board members. This captures an organization's total human resources, not limited to paid staff. Prior research shows that paid (wage) and flexible labor—such as independent contractors and freelancers—are not entirely separate resources but substitutable ones within nonprofit arts organizations (Woronkowicz et al., 2020), suggesting that different labor types function as parts of a single pool of human resources; on this basis, volunteers, interns, and board members are likewise included to capture an organization's total capacity to mobilize human resources. A log transformation was applied to correct the right-skewed distribution of staffing size.

Programmatic capacity (H2) is measured as the natural log of total program occurrences (ln_programs), corresponding to the program design and operation component of Hall et al.'s (2003) structural capacity. The variable total_program_occurrences sums the number of performances, school programs, community programs, and other activities. Running a large number of programs requires capacity along several dimensions—physical infrastructure and equipment, staff skill and creative capacity, and the operational systems needed to plan and deliver programming. Therefore, an organization that delivers more programs is presumed to possess a correspondingly greater capacity to design, plan, and operate them. This treatment is consistent with Kim (2017), who similarly relies on the log-transformed number of performances as a program-related indicator among U.S. nonprofit performing arts organizations using the same CDP data source, finding it significantly associated with organizational outcomes.

Financial capacity (H3) is measured as months of operating cash on hand (months_of_operating_cash_total). The scholarly tradition of measuring financial capacity traces back to Tuckman and Chang's (1991) concept of financial vulnerability, which holds that an organization's flexibility in responding to financial shocks depends on the availability of financial slack. Bowman (2011) extends this concept, defining nonprofit financial capacity as consisting of "resources that give an organization the wherewithal to seize opportunities and react to unexpected threats," and proposes months of spending—expendable, unrestricted resources relative to operating expenses—as its core short-term indicator. The CDP's months of operating cash measure operationalizes this concept. This measure represents the length of time an organization could continue operating without external income, reflecting the financial capacity to invest in new revenue-generating activities and bear the associated risk. Accordingly, greater financial capacity is expected to be positively associated with the earned revenue ratio. Because it is a ratio variable, it is independent of organizational size.

The remaining subdimensions of structural capacity proposed by Hall et al. (2003) are excluded from this analysis. Relational and network capacity is difficult to operationalize using financial data, and infrastructure capacity (such as facility ownership) shows little variation over time, making it unidentifiable within a fixed-effects framework.

As a control variable, the natural log of contributed revenue (ln_contrib) is included. Because the dependent variable is a ratio, an increase in contributed revenue mechanically enlarges the denominator, producing a negative relationship with the earned revenue share. Without controlling for this structural relationship, estimates could be biased; contributed revenue is therefore included as a control variable.

3-3. Analytical Approach

This study employs a two-way fixed effects (TWFE) panel regression as its primary analytical method (Wooldridge, 2010), specified as:

$$Y\_it = \beta 1\, ln(positions)\_it + \beta 2\, ln(programs)\_it + \beta 3\, (months\_cash)\_it + \beta 4\, ln(contrib)\_it + \alpha\_i + \gamma\_t + \varepsilon\_it$$

where Y_it denotes the earned revenue ratio of organization i in year t; α_i and γ_t represent organization and year fixed effects, respectively; and ε_it is the idiosyncratic error term, clustered at the organization level. Organization fixed effects control for time-invariant characteristics such as founding history, geographic location, and genre, allowing estimation to focus on within-organization change over time. Year fixed effects control for shocks common across organizations in a given year, such as the pandemic, inflation, and policy changes. Standard errors are clustered at the organization level. All analyses were conducted using the fixest package (feols function) in R.

Because ln_positions, ln_programs, and months_cash are measured in different units, their coefficients are not directly comparable in magnitude. To assess whether human and programmatic capacity are associated with the earned revenue ratio to different degrees, a Wald test of coefficient equality is conducted, testing the null hypothesis that β_ln_positions = β_ln_programs. In addition, standardized effect sizes are calculated by multiplying each coefficient by the standard deviation of the corresponding variable, expressing the association of each capacity dimension in a common metric (percentage-point change in the earned revenue ratio per one-standard-deviation increase).

Estimates from the TWFE model reflect associations rather than causal effects, and the possibility of reverse causality cannot be fully ruled out; this is acknowledged as a key limitation of the study. To assess the sensitivity of the results, three robustness checks are conducted: a model excluding fiscal years 2020–2021, when the arts and cultural sector faced its most severe pandemic disruption (Betzler et al., 2021); a model additionally controlling for organizational scale (total assets); and a model using the natural log of earned revenue as an alternative dependent variable (see Section 4.3 and Appendix Table A1).

Results

4-1. Descriptive Statistics

The estimation sample consists of 7,489 organization-year observations from 1,864 nonprofit performing arts organizations. By genre, the sample is distributed as follows: Music (2,578, 34.4%), Theater (2,250, 30.0%), General Performing Arts (1,051, 14.0%), Dance/Ballet (983, 13.1%), and Opera (627, 8.4%).

Table 1 presents descriptive statistics for the main variables. The dependent variable, earned revenue ratio, has a mean of 0.333 (SD = 0.235), indicating that organizations in the sample cover, on average, approximately 33% of total operating revenue through earned income. The median (0.290) is lower than the mean, reflecting a right-skewed distribution. For staffing size, the median is 80 while the mean is 165—more than twice as high—and for program occurrences, the median is 36 while the mean reaches 195. For the financial capacity variable, months_cash, the median is 3.3 months, indicating that half of the organizations in the sample held cash reserves sufficient to cover no more than about three months of operating expenses.

Table 1. Descriptive Statistics

Variable	**N**	**Mean**	**SD**	**Min**	**Median**	**Max**
Earned Revenue Ratio	7,489	0.33	0.23	0	0.29	1.00
Staff Positions	7,489	164.50	618.92	0	80.00	48,500.00
Program Occurrences	7,489	195.33	521.06	0	36.00	5,832.00
Months of Cash	7,489	5.08	5.71	0	3.32	36.52
Contributed Revenue ($)	7,489	1,297,547.31	5,263,334.90	0	186,408.00	190,422,000.00
Organization Age (yrs)	7,139	34.74	25.52	0	29.00	165.00

Note: Statistics are based on the estimation sample (N = 7,489); organization age is missing for 350 observations. Sample period: FY2019–FY2024.

4-2. Main Results

Table 2 presents the results of the two-way fixed effects regression. The model's Within R^2 is 0.325, indicating that the model's explanatory variables account for roughly a third of the year-to-year variation in the earned revenue ratio within the same organization. The Adjusted R^2 of 0.798 reflects the explanatory power of the full model, including the organization and year fixed effects. Variance inflation factors (VIF)—a diagnostic for whether the independent variables are too closely related to one another to be reliably distinguished—ranged from 1.0 to 1.9 in the main model, and remained below conventional thresholds (maximum 3.2) in the size-controlled specification, indicating no evidence of problematic multicollinearity.

With respect to H1, ln_positions showed a significant positive relationship with the earned revenue ratio ($\beta = 0.048$, SE = 0.004, $p < 0.001$). For example, holding other conditions constant, an organization with 100 staff positions is estimated to have an earned revenue ratio approximately 1.1 percentage points higher than an otherwise comparable organization with 80 positions.

With respect to H2, ln_programs also showed a significant positive relationship with the earned revenue ratio ($\beta = 0.022$, SE = 0.002, $p < 0.001$). For example, holding other conditions constant, an organization with 50 program occurrences is estimated to have an earned revenue ratio approximately 0.7 percentage points higher than an otherwise comparable organization with 36 occurrences. Human capacity exhibited a significantly stronger association than programmatic capacity (see Table 2 note for the formal test of coefficient equality).

Contrary to H3, months_cash shows a statistically significant negative association with the earned revenue ratio ($\beta = -0.0037$, $p < .001$). A one-unit increase in months of cash on hand is estimated to be associated with an approximately 0.4

percentage point decrease in the earned revenue ratio. This finding runs counter to the hypothesized direction and stands in contrast to the positive associations found for human and programmatic capacity, indicating that the three capacity dimensions are not associated with earned revenue in a uniform direction. Possible explanations for this reversal are discussed in Section 5.

This ranking is robust to differences in variable scale: a one-standard-deviation increase in ln_positions is associated with a 5.1 percentage-point increase in the earned revenue ratio, compared with 3.8 percentage points for ln_programs and a 2.1 percentage-point decrease for months_cash. It should be noted, however, that these standardized effects are modest in absolute magnitude; the focus of this study is not the size of these effects but the relative ranking among the three capacity dimensions.

Table 2. Main Results: Two-Way Fixed Effects Regression

Variable	**Coefficient**	**Std. Error**	**Effect per 1 SD (pp)**
Human capacity (ln_positions)	0.0478***	(0.0043)	5.12
Programmatic capacity (ln_programs)	0.0219***	(0.0019)	3.80
Financial capacity (months_cash)	-0.0037***	(0.0005)	-2.10
Contributed revenue (ln_contrib)	-0.1080***	(0.0078)	—
Organization Fixed Effects	Yes		
Year Fixed Effects	Yes		
Observations	7,489		
Within R^2	0.325		

Variable	Coefficient	Std. Error	Effect per 1 SD (pp)
Adjusted R^2	0.798		

*Note: Dependent variable is earned revenue ratio. Standard errors clustered at the organization level in parentheses. A Wald test of coefficient equality between ln_positions and ln_programs rejects the null of equal coefficients ($z = 5.52$, $p < .001$). "Effect per 1 SD" reports the change in the earned revenue ratio (in percentage points) associated with a one-standard-deviation increase in each capacity variable. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.*

4-3. Robustness Checks

Table 3 presents the results of two robustness checks. First, in a model excluding fiscal years 2020 and 2021—when performing arts organizations' operations were most severely constrained by the COVID-19 pandemic—the direction and significance of H1 ($\beta = 0.030$, $p < 0.001$), H2 ($\beta = 0.018$, $p < 0.001$), and H3 ($\beta = -0.002$, $p < 0.001$) remained consistent. The pattern of human capacity showing a stronger association than programmatic capacity also held. The smaller magnitude of the human capacity coefficient in this subsample suggests that pandemic-era variation amplified the main estimate to some degree; the excluded-years estimate may therefore be read as a conservative bound, with the substantive conclusions unchanged.

Second, in a model additionally controlling for organizational scale (log total assets, ln_assets), the coefficient for ln_positions decreased modestly from 0.048 to 0.043 but remained fully significant ($p < 0.001$). This model includes fewer observations (6,401) because total assets are not reported by all organizations. months_cash also remained significant after controlling for scale, revealing the H3 pattern more clearly. This indicates that neither the human capacity effect nor the financial capacity effect is simply explained by organizational scale.

Finally, because the dependent variable is a ratio whose denominator includes contributed revenue, the results could in principle reflect the mechanical relationship between the ratio's components rather than substantive associations. To address this concern, an alternative specification uses the natural log of earned revenue as the dependent variable, controlling for total operating revenue (Appendix Table A1). All three capacity variables retain their sign and significance, and human capacity again shows a larger coefficient than programmatic capacity, indicating that the findings do not depend on the ratio-based construction of the dependent variable. Notably, the negative association of months_cash persists even when earned revenue is measured in levels, suggesting that this pattern is not an artifact of revenue composition.

Table 3. Robustness Checks

Variable	**Main**	**Excl. 2020-2021**	**Size Controlled**
Human capacity (ln_positions)	0.0478*** (0.0043)	0.0299*** (0.0053)	0.0433*** (0.0043)
Programmatic capacity (ln_programs)	0.0219*** (0.0019)	0.0183*** (0.0028)	0.0186*** (0.0018)
Financial capacity (months_cash)	-0.0037*** (0.0005)	-0.0022*** (0.0006)	-0.0058*** (0.0005)
Contributed revenue (ln_contrib)	-0.1080*** (0.0078)	-0.1042*** (0.0097)	-0.1318*** (0.0099)
Organizational scale (ln_assets)			0.0121*** (0.0027)
Observations	7,489	4,347	6,401
Within R^2	0.325	0.311	0.364

Variable	Main	Excl. 2020-2021	Size Controlled

Note: Dependent variable is earned revenue ratio. All models include organization and year fixed effects; standard errors clustered at the organization level in parentheses. "Excl. 2020-2021" excludes fiscal years during which the COVID-19 pandemic most severely disrupted performing arts operations. "Size Controlled" adds log total assets (ln_assets) as a control for organizational scale. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Discussion and Conclusion

5-1. Theoretical Implications of the Main Findings

The results indicate that the three capacity dimensions are not associated with earned revenue in a uniform way: human and programmatic capacity show positive associations, with human capacity the stronger of the two, while financial capacity shows a negative pattern. The following discussion considers each hypothesis in relation to its corresponding literature.

The finding that human capacity is positively associated with earned revenue is consistent with Hall et al.'s (2003) characterization of human capital as "the key element that leads to the development of all other capacities." This association also connects to a structural feature of the performing arts documented in cultural economics. As Baumol and Bowen's (1966) cost disease thesis established, the performing arts are inherently labor-intensive: unlike manufacturing, no substitution of capital for labor can reduce the number of people required to stage a performance. A similar logic may also apply to revenue generation. Ticket sales do not arise simply from producing a performance; they also depend on marketing, audience development, educational outreach, and relationship-building with patrons—activities carried out primarily through human resources rather than capital alone. In this light, the labor-intensive character of both production and revenue generation may explain why earned revenue is tied more closely to human capacity than to the other dimensions.

Programmatic capacity was operationalized as the total number of program occurrences—a measure of programming volume. Although this measure was positively associated with earned revenue, the association was weaker than that observed for human capacity, suggesting that the sheer volume of programming may be a less powerful lever for expanding earned revenue than the scale of human capacity. This is consistent with the benefits theory of nonprofit finance: Wilsker and Young (2010) show that it is the composition of an organization's programs—not merely their number—that shapes its revenue-source structure, with activities in which beneficiaries pay directly for services associated with greater reliance on earned income. Liu and Kim (2022) extend this finding using a larger CDP-based sample of arts and cultural nonprofits, showing that organizations whose programs confer largely private, excludable benefits—such as performing arts organizations—experience improved financial health as their program-based (private) revenue share increases, once it surpasses a critical threshold. Read together, this literature suggests that the composition of an organization's programming—specifically, the balance between activities with private, excludable benefits (such as ticketed performances) and those with collective, non-excludable benefits (such as free community or school programs)—may matter as much as, or more than, program volume in determining how effectively programmatic activity translates into earned revenue.

The most unexpected finding concerns financial capacity. H3 predicted a positive association, on the premise that financial slack enables investment in earned-revenue generation; the data instead reveal a significant negative association. Two interpretations, which are not mutually exclusive, merit consideration. The first is behavioral: organizations maintaining relatively larger cash reserves may be making a conservative allocation choice, retaining financial slack rather than deploying it toward activities associated with earned revenue generation, such as staffing or program development. This interpretation is broadly consistent with Bowman's (2011) definition of financial capacity as the resources that enable organizations to seize opportunities and respond to threats—resources that, in this reading, are being held rather than used. The second interpretation is mechanical: because the fixed effects model is identified from year-to-year variation within organizations, temporary spikes in cash holdings—such as those following a major grant, bequest, or capital campaign—could coincide with years in which contributed revenue is unusually high relative to earned revenue, producing a negative within-organization association that reflects the timing of fundraising cycles rather than a substantive trade-off. That the negative association persists when earned revenue is measured in levels rather than as a share (Appendix Table A1) weighs against a purely compositional account, but does not rule out timing effects. Because this study does not observe how financial reserves are actually deployed, distinguishing between these interpretations—for example, by examining the sources and subsequent uses of cash inflows—remains a task for future research. More broadly, the results extend Gregory and Howard's (2009) resource-constraint perspective by suggesting that nonprofit organizations may face trade-offs in allocating limited resources across different forms of organizational capacity.

Taken together, these findings suggest that organizational capacity should not be treated as a single, unified construct when examining earned revenue generation. The contribution of this study is therefore not to argue that capacity matters, but to show that capacity dimensions matter unequally, and to provide one of the first empirical basis for comparing them within a single model.

5-2. Policy Implications

These findings speak to how investment priorities among organizational capacities might be set when policy resources are limited. The capacity most closely associated with earned revenue was human capacity, not programmatic capacity—suggesting that cultural policy may need to consider more explicitly which capacity to prioritize, rather than treating capacity support as an undifferentiated goal.

Public support is often structured around individual projects. Because project-based funding is typically restricted to specific activities, it may do little to strengthen the kind of long-term human capacity that organizations need internally—leaving organizations with little room to plan for stable employment or accumulate long-term expertise. This interpretation aligns with the qualitative research underlying Hall et al.'s (2003) capacity framework: nonprofit organizations in Canada participating in that study widely reported that funders often provide project or service contracts rather than support for core operations, leaving them with persistent difficulty recruiting and retaining staff and engaging in long-term planning. The present finding—that human capacity shows a stronger association with earned revenue than programmatic capacity—raises the possibility that this funding structure does not necessarily align with the capacity that underpins organizations' long-term financial sustainability. In this context, funding approaches such as general operating support and multi-year funding may warrant greater attention. Arts Council England's National Portfolio Organisation program offers a working example, providing multi-year funding directed at an organization's core operations rather than individual projects. This study does not test the effects of such instruments, but they represent policy mechanisms worth evaluating in future research.

With respect to programmatic capacity, the present findings do not suggest that project-based funding lacks value. As discussed in Section 5.1, the comparatively modest association observed for programmatic capacity may reflect the composition of programming—the balance between fee-based and collective-benefit activities—as much as its volume, and may also depend on the human capacity needed to convert programming into revenue through marketing and audience engagement. To the extent that program composition matters, funding mechanisms that are sensitive mainly to the volume of programming delivered—such as per-program or per-performance support—are unlikely to fully capture what enables organizations to convert programming into earned revenue. Funders may therefore wish to consider, alongside the amount of programming supported, the kind of programming being funded. Neither interpretation was directly tested in this study, and future research incorporating measures of program composition or audience engagement would help clarify which mechanism, or combination of mechanisms, is at work.

The reversed pattern observed for financial capacity (H3) likewise offers a starting point for policy discussion. If high levels of cash reserves partly reflect underinvestment in human or programmatic capacity, this raises the question of whether existing policy indicators for assessing financial health are sufficient. This concern is consistent with prior evidence that financial indicators do not uniformly predict favorable outcomes for nonprofit arts organizations: using CDP data, Kim (2017) finds that financial attributes conventionally assumed to strengthen fiscal health are not always associated with better program outcomes, and Kirchner et al. (2007), studying U.S. symphony orchestras, similarly find a significant negative relationship between the level of government support and financial health, suggesting that external support does not necessarily translate directly into an organization's financial stability. This does not imply that holding large cash reserves is undesirable, but rather points to a need to consider whether that financial slack is actually being deployed toward strengthening organizational capacity.

These policy implications should not be interpreted as evidence that any specific funding instrument is effective. Rather, this study identifies which dimension of organizational capacity is most closely associated with earned revenue, thereby providing empirical grounds for future research to evaluate whether instruments such as general operating support, multi-year funding, or workforce development programs are effective in strengthening human capacity.

5-3. Limitations

This study has several limitations. First, although the two-way fixed effects model controls for unobserved time-invariant heterogeneity, it does not establish causal relationships. In particular, the possibility of reverse causality—that stronger earned revenue may itself facilitate the development of organizational capacity—cannot be ruled out.

Second, the measures of organizational capacity inevitably simplify complex organizational constructs, and they share a common limitation: each captures the quantity of a resource but not its quality. Human capacity is measured as total workforce size, which does not reflect leadership, professional expertise, or managerial skill—dimensions that Hall et al. (2003) include within human resources capacity. Programmatic capacity is measured as the number of program occurrences, which does not reflect the quality or composition of programming, such as the balance between fee-based and free offerings discussed in Section 5.1. Financial capacity, likewise, is measured by months of cash on hand, a single dimension of a broader construct. One specific concern—that workforce size may partly reflect organizational scale rather than capacity—is mitigated by the robustness check controlling for total assets, but the broader point stands: future research employing richer measures of staff expertise, program composition, and financial structure could refine the comparisons reported here.

Third, this study identifies which dimensions of organizational capacity are associated with earned revenue but does not examine which specific policy interventions most effectively strengthen those capacities. Future research should therefore investigate whether different funding mechanisms—such as general operating support, multi-year funding, or workforce development programs—produce measurable improvements in organizational capacity.

Fourth, because CDP data are based on voluntary reporting, smaller organizations may be underrepresented. Organizations participating in the CDP may possess relatively stronger administrative capacity than the broader population of nonprofit performing arts organizations, which may limit the generalizability of the findings.

Finally, this study examines the performing arts sector as a whole. Because revenue structures and organizational practices may differ across artistic disciplines, future research could explore whether the relative importance of different capacity dimensions varies across genres.

5-4. Conclusion

Organizational capacity is widely accepted as important for the financial sustainability of nonprofit performing arts organizations, but capacity is not a single thing, and funders cannot invest in all of its dimensions at once. This study asked which dimension—human, programmatic, or financial—is most closely associated with earned revenue, and answered it by placing all three within a single fixed-effects framework using national panel data.

The finding that human capacity shows the strongest association with earned revenue among the three dimensions suggests that capacity dimensions may matter unequally rather than interchangeably. This is not intended to prescribe a specific policy instrument; the finding identifies where the strongest association lies, not which funding mechanism best strengthens it.

This contribution should be read alongside the limitations discussed above. The associations reported here are not causal, the human and financial capacity measures carry their own measurement limitations, and the study does not directly test which funding approaches actually strengthen human capacity or whether such changes would, in turn, increase earned revenue. Confirming these mechanisms remains a task for future policy experiments, quasi-experimental research, or long-term longitudinal studies, and future work extending this comparison across genres or using more precise measures of capacity would further clarify how robust this hierarchy is.

Nonetheless, by shifting the question from whether capacity matters to which capacity matters most, this study offers an empirical starting point for future work on capacity investment priorities: if subsequent research establishes that these associations reflect causal relationships, the findings reported here would indicate where such investment is most likely to pay off.

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organizational data, and results are reported only in aggregated form.

Disclosure statement

The author report there are no competing interests to declare.

Declaration of generative AI use

The author used Claude (Anthropic, Claude Sonnet 5) for language editing and manuscript structuring. The author has reviewed this tool's terms of use and confirms its suitability for publication, and confirms that the originality and accuracy of all content—including references—have been verified, taking full responsibility for the integrity of the manuscript.

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Appendix

Table A1. Alternative Dependent Variable: ln(Earned Revenue)

Variable	**Coefficient**	**Std. Error**
Human capacity (ln_positions)	0.2570***	(0.0432)
Programmatic capacity (ln_programs)	0.1401***	(0.0206)
Financial capacity (months_cash)	-0.0288***	(0.0057)
Total operating revenue (ln_totrev)	1.0619***	(0.0775)
Organization Fixed Effects	Yes	

Variable	Coefficient	Std. Error
Year Fixed Effects	Yes	
Observations	7,489	
Within R^2	0.201	

*Note: Dependent variable is the natural log of earned revenue (plus 1). Total operating revenue (log) is included to control for organizational scale. Standard errors clustered at the organization level in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.*